
AI Server Benchmark Performance Report

Run ID: run_sim_20260506_demo

Concurrency Scaling • Model: llama3.2-deterministic:latest
2026-05-06 11:13

Servers Evaluated: 2

Total Test Cases: 120

Prompts Processed: 7271

Duration: 3600.0s

Generated by **aiDaptive Benchmark Suite v2.1**
2026-05-06 11:13

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1 Executive Summary

This report presents the benchmark performance analysis for run run_sim_20260506_demo, executed on 2026-05-06 11:13. The benchmark evaluated 2 server(s) using the **Concurrency Scaling** test suite with the **llama3.2-deterministic:latest** model.

1.1 Test Configuration

Parameter	Value
Test Suite	Concurrency Scaling
Model	llama3.2-deterministic:latest
Servers	34.142.222.133, 35.186.159.250
Tools Used	litellm, oha, ollama_native
Scenarios	code_generation, long_context, reasoning, simple_chat, translation
Concurrency Levels	
Total Prompts	7271
Start Time	2026-05-06 08:00:00
End Time	2026-05-06 09:00:00
Duration	3600.0s

1.2 Overall Winner

Based on aggregate performance metrics across all test scenarios, **35.186.159.250** demonstrated superior performance.

2 Performance Analysis

The following charts show key performance metrics across different concurrency levels. Each metric is measured at p50, p95, and p99 percentiles to provide a complete distribution view.

2.1 Time to First Token (TTFT)

TTFT measures the latency from request submission to receiving the first token. Lower values indicate faster initial response.

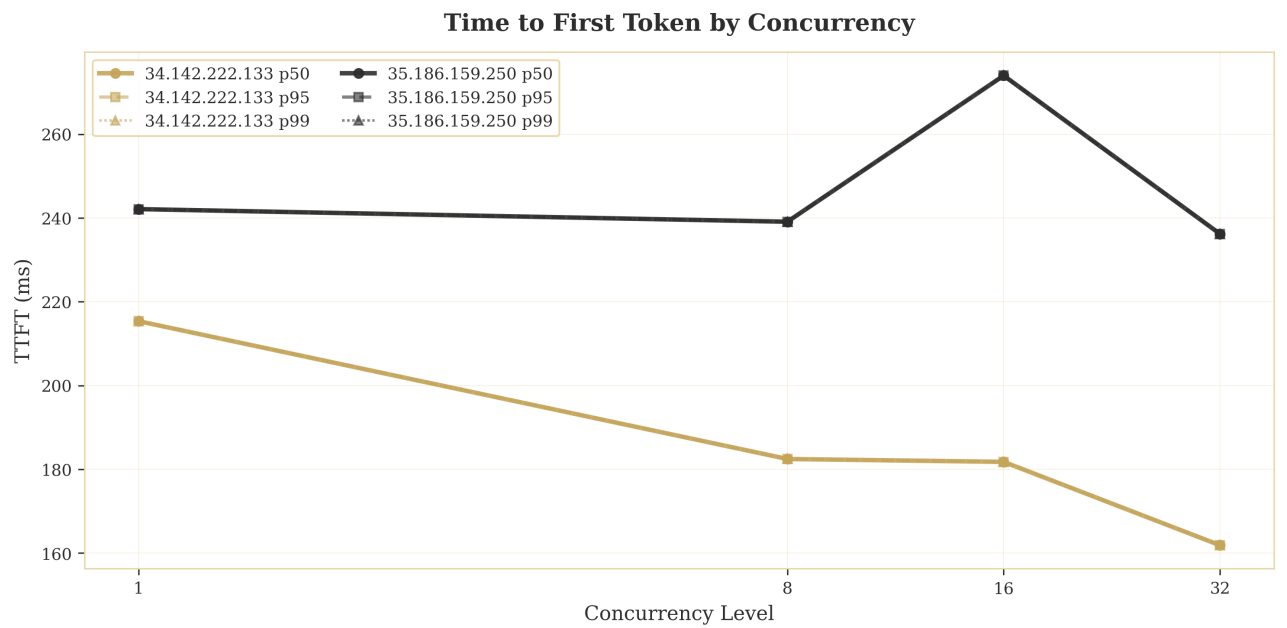


Figure 1: TTFT by Concurrency Level (p50/p95/p99)

2.2 Inter-Token Latency (ITL)

ITL measures the average time between consecutive tokens during generation. Lower values indicate smoother streaming.

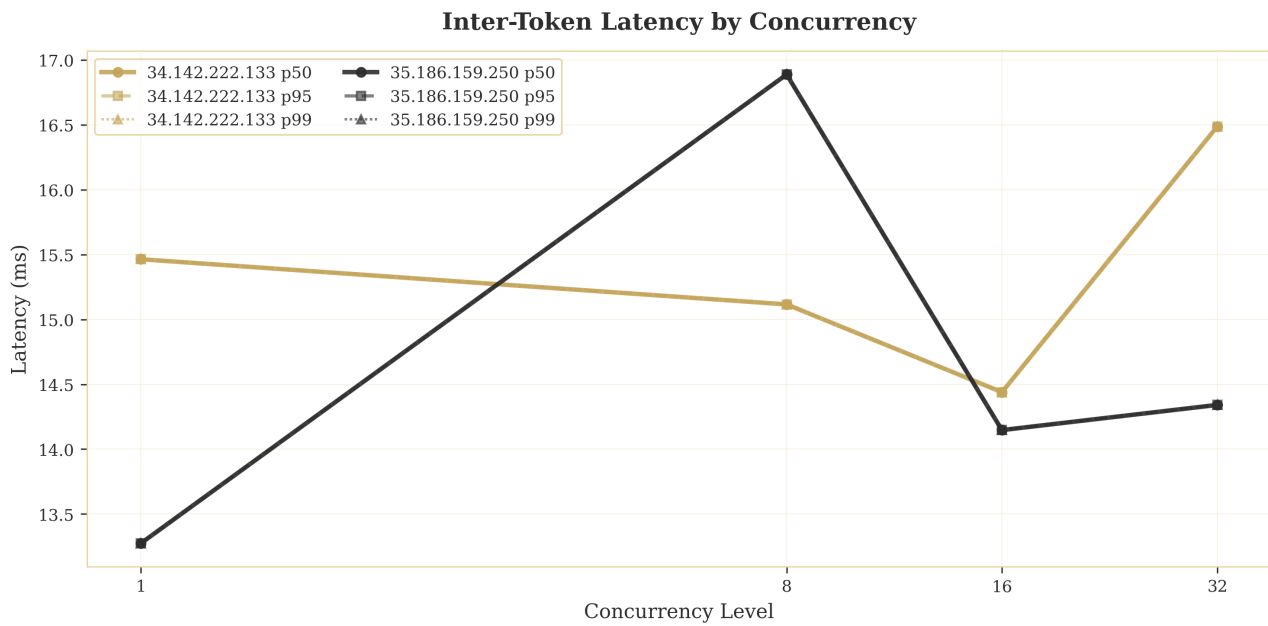


Figure 2: Inter-Token Latency by Concurrency Level

2.3 Tokens Per Second Per User (TPS)

TPS measures the throughput experienced by each individual user. Higher values indicate better throughput efficiency.

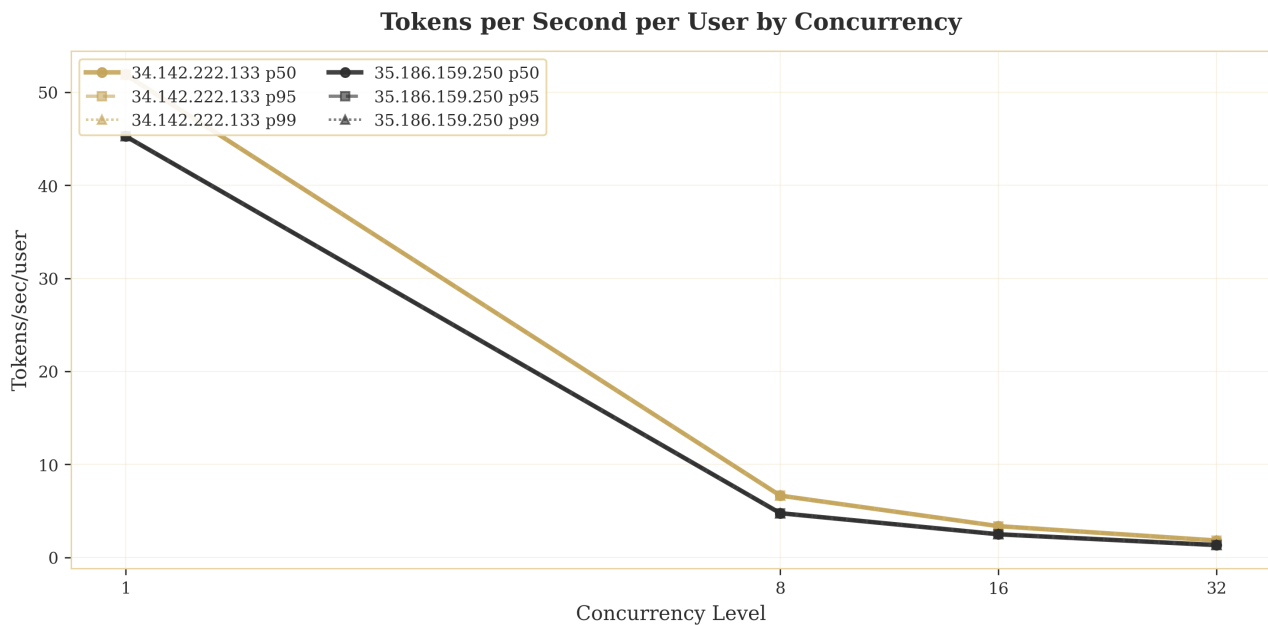


Figure 3: Tokens/sec/user by Concurrency Level

2.4 Request Latency

End-to-end request latency from submission to completion.

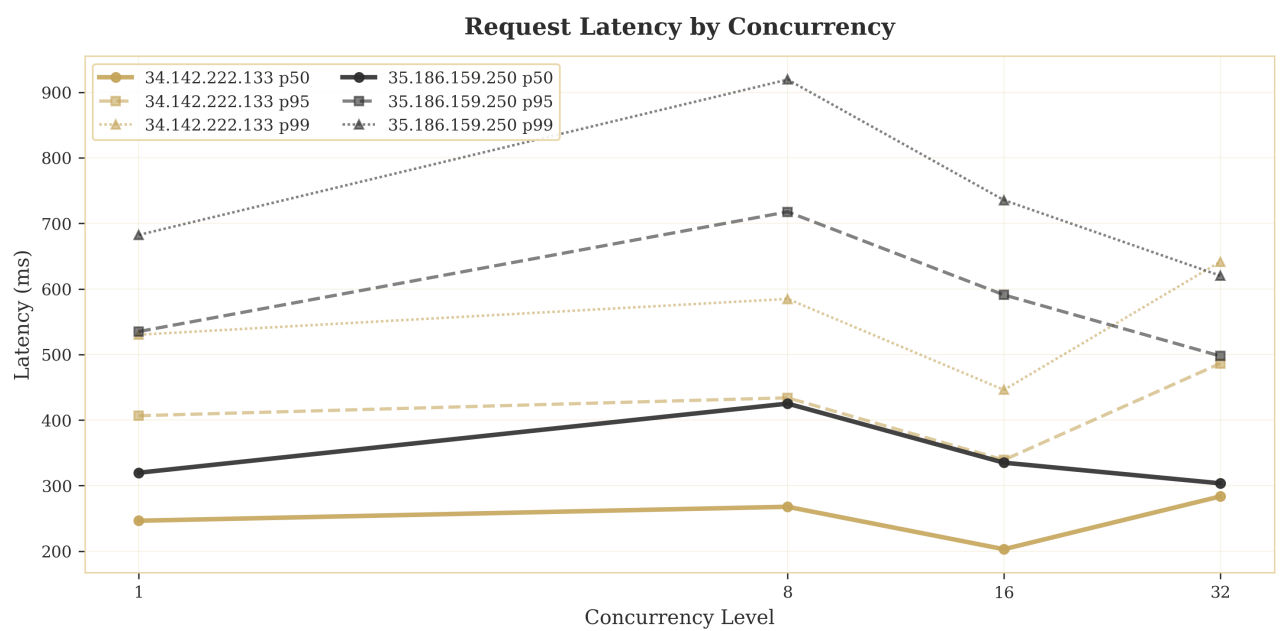


Figure 4: Request Latency by Concurrency Level (p50/p95/p99)

3 Server Comparison

3.1 Side-by-Side Comparison

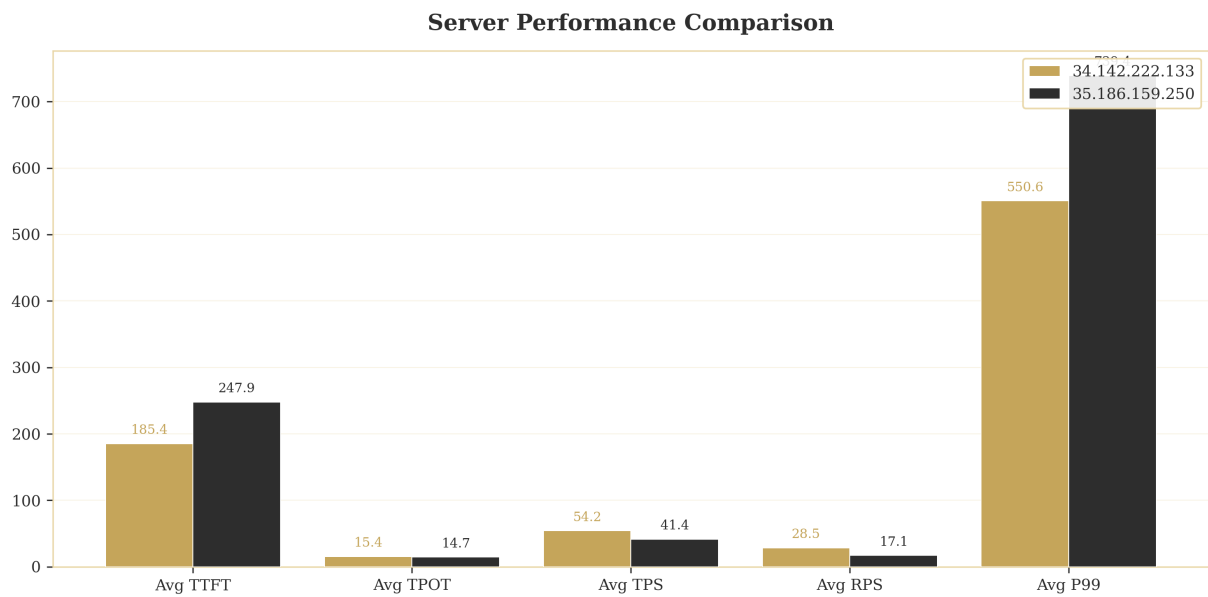


Figure 5: Average Performance Metrics by Server

3.2 Delta Analysis

The heatmap below shows the percentage difference between servers for each test configuration. Green indicates improvement, red indicates degradation.

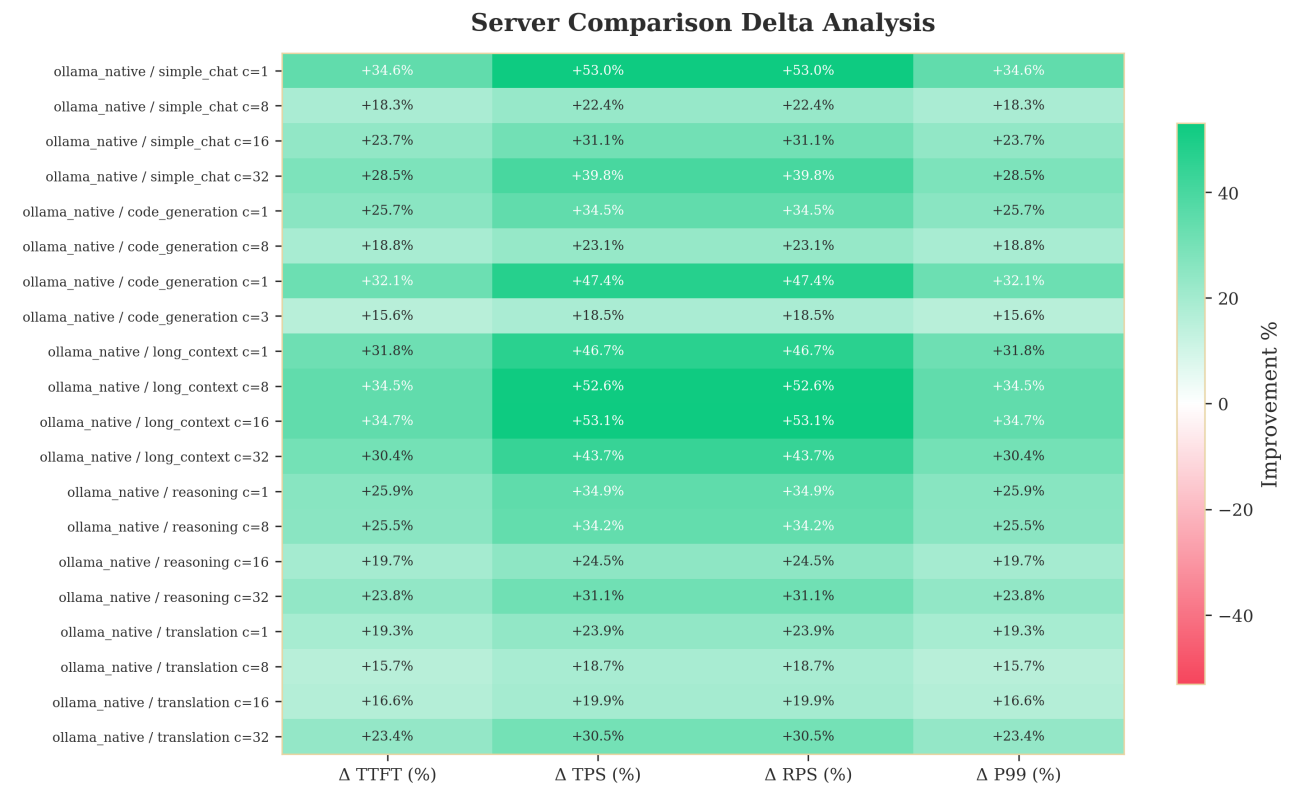


Figure 6: Server Comparison Delta Heatmap

3.3 Detailed Comparison Table

Tool	Scenario	C	Server 1				Server 2				Win
			TTFT	TPS	RPS	P99	TTFT	TPS	RPS	P99	
ollama_native	simple_chat	1	205.28	35.99	14.46	492.41	134.19	55.06	22.11	321.90	ser
ollama_native	simple_chat	8	372.58	25.83	6.42	536.37	304.36	31.62	7.86	438.16	ser
ollama_native	simple_chat	16	394.38	26.07	7.67	1020.26	300.94	34.16	10.06	778.52	ser
ollama_native	simple_chat	32	391.25	27.85	5.20	853.99	279.84	38.94	7.27	610.82	ser
ollama_native	code_generation	1	256.89	30.96	12.89	1052.29	190.92	41.65	17.34	782.09	ser
ollama_native	code_generation	8	356.49	49.82	3.97	896.53	289.60	61.33	4.89	728.32	ser
ollama_native	code_generation	16	172.88	23.87	9.98	593.43	117.29	35.18	14.71	402.63	ser
ollama_native	code_generation	32	359.31	33.23	7.52	640.45	303.21	39.38	8.91	540.46	ser
ollama_native	long_context	1	294.44	18.22	12.32	453.60	200.74	26.73	18.07	309.24	ser
ollama_native	long_context	8	325.94	21.70	13.69	887.13	213.59	33.12	20.90	581.33	ser
ollama_native	long_context	16	175.11	23.90	8.64	504.29	114.40	36.58	13.22	329.47	ser
ollama_native	long_context	32	152.66	30.64	10.08	907.57	106.23	44.03	14.49	631.52	ser
ollama_native	reasoning	1	323.35	46.35	12.86	470.52	239.70	62.53	17.35	348.79	ser
ollama_native	reasoning	8	226.58	21.07	6.43	880.86	168.85	28.28	8.63	656.44	ser
ollama_native	reasoning	16	357.65	17.02	10.87	753.91	287.18	21.20	13.54	605.37	ser
ollama_native	reasoning	32	188.63	28.43	5.59	1082.92	143.83	37.29	7.33	825.72	ser
ollama_native	translation	1	384.68	49.36	4.54	593.36	310.60	61.13	5.62	479.09	ser
ollama_native	translation	8	335.49	47.19	10.28	661.36	282.74	55.99	12.19	557.37	ser
ollama_native	translation	16	347.35	40.87	3.88	645.47	289.77	48.99	4.66	538.46	ser
ollama_native	translation	32	164.16	44.65	3.97	629.83	125.81	58.26	5.18	482.69	ser
oha	simple_chat	1	276.24	41.07	5.62	713.54	206.05	55.06	7.54	532.26	ser
oha	simple_chat	8	247.78	42.27	13.75	545.78	196.86	53.21	17.31	433.61	ser

Tool	Scenario	C	Server 1				Server 2				Win
			TTFT	TPS	RPS	P99	TTFT	TPS	RPS	P99	
oha	simple_chat	16	228.91	19.63	13.94	444.55	187.47	23.98	17.02	364.06	ser
oha	simple_chat	32	157.58	42.97	3.07	699.83	130.06	52.07	3.73	577.61	ser
oha	code_generation	1	197.95	45.64	7.03	478.32	131.37	68.78	10.59	317.43	ser
oha	code_generation	8	223.85	32.63	7.29	725.60	174.57	41.85	9.35	565.85	ser
oha	code_generation	16	361.15	36.99	7.31	624.68	236.86	56.40	11.14	409.70	ser
oha	code_generation	32	206.17	28.62	14.90	942.26	174.52	33.81	17.61	797.58	ser
oha	long_context	1	269.46	46.61	12.98	846.59	227.10	55.30	15.40	713.49	ser
oha	long_context	8	213.83	22.76	5.58	765.60	160.37	30.34	7.44	574.21	ser
oha	long_context	16	251.22	49.13	8.34	1070.47	197.21	62.58	10.63	840.33	ser
oha	long_context	32	196.08	23.74	14.37	686.31	150.19	30.99	18.76	525.69	ser
oha	reasoning	1	158.98	15.48	10.36	904.82	107.75	22.84	15.28	613.24	ser
oha	reasoning	8	263.13	22.38	11.30	1005.05	207.68	28.35	14.32	793.26	ser
oha	reasoning	16	310.78	29.37	9.52	1164.20	202.38	45.10	14.62	758.13	ser
oha	reasoning	32	209.88	31.98	12.66	1100.77	163.38	41.09	16.26	856.88	ser
oha	translation	1	163.82	30.27	14.93	723.77	125.46	39.52	19.50	554.32	ser
oha	translation	8	385.98	49.43	11.81	432.70	287.88	66.27	15.84	322.72	ser
oha	translation	16	155.67	46.68	4.22	623.05	114.81	63.29	5.72	459.52	ser
oha	translation	32	373.85	33.63	6.26	1016.00	287.30	43.75	8.14	780.81	ser
litellm	simple_chat	1	248.70	46.56	3.02	1053.01	184.90	62.63	4.06	782.91	ser
litellm	simple_chat	8	173.46	25.22	4.29	917.85	140.61	31.12	5.29	744.01	ser
litellm	simple_chat	16	265.95	20.08	14.81	633.54	205.98	25.93	19.12	490.68	ser
litellm	simple_chat	32	278.60	41.34	9.05	983.64	182.45	63.13	13.82	644.15	ser
litellm	code_generation	1	291.05	20.97	5.62	735.78	217.16	28.11	7.54	548.98	ser
litellm	code_generation	8	241.60	26.44	10.80	913.14	179.32	35.62	14.55	677.73	ser
litellm	code_generation	16	335.14	46.89	6.00	1061.49	226.30	69.45	8.88	716.76	ser
litellm	code_generation	32	241.58	41.46	3.78	564.46	178.53	56.10	5.11	417.15	ser
litellm	long_context	1	398.05	46.14	12.25	777.47	272.30	67.45	17.91	531.85	ser
litellm	long_context	8	377.71	43.16	9.82	1005.15	280.64	58.08	13.22	746.83	ser
litellm	long_context	16	162.46	34.96	14.98	445.66	125.33	45.31	19.42	343.80	ser
litellm	long_context	32	360.17	19.73	14.49	869.86	272.94	26.03	19.12	659.18	ser
litellm	reasoning	1	382.08	22.36	14.31	857.38	282.65	30.22	19.35	634.26	ser
litellm	reasoning	8	328.17	24.11	5.14	1009.86	236.93	33.40	7.12	729.09	ser
litellm	reasoning	16	213.89	28.90	5.25	930.79	141.73	43.61	7.92	616.80	ser
litellm	reasoning	32	376.31	22.67	12.56	1137.81	292.53	29.16	16.16	884.50	ser
litellm	translation	1	310.53	20.97	3.49	963.35	260.15	25.03	4.17	807.07	ser
litellm	translation	8	337.01	31.72	4.10	1066.34	227.26	47.04	6.08	719.10	ser
litellm	translation	16	172.86	42.62	5.70	702.91	137.37	53.64	7.17	558.58	ser
litellm	translation	32	305.31	37.66	5.76	728.81	248.01	46.36	7.09	592.04	ser

4 Hardware Utilization

4.1 Hardware Specifications

Metric	34.142.222.133	35.186.159.250
GPU		
GPU Model	NVIDIA T4	NVIDIA L4
GPU Util (avg)	57.47	76.53
GPU Util (max)	74.7	94.8
VRAM Used (avg)	15.01 GB	14.39 GB
VRAM Total	16.00 GB	24.00 GB
Power (avg)	196.4 W	203.2 W
Power (max)	276.7 W	280.0 W
Temp (avg)	68.45	68.05
Temp (max)	81.8	81.1
CPU / RAM		
CPU (avg)	50.71	49.34
CPU (max)	69.9	69.7
RAM Used (avg)	10.11 GB	10.15 GB
RAM Total	16.00 GB	32.00 GB
Disk / Network		
Disk Read (avg)	114.89 MB/s	103.50 MB/s
Disk Write (avg)	47.22 MB/s	50.64 MB/s

4.2 Resource Utilization Timeline

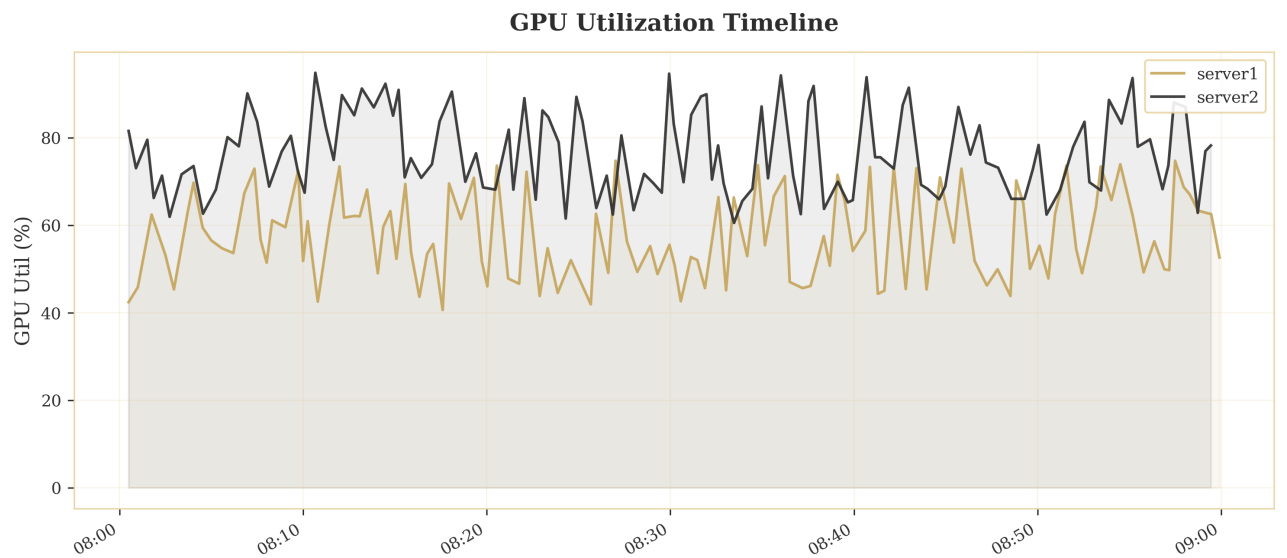


Figure 7: GPU Utilization Over Time

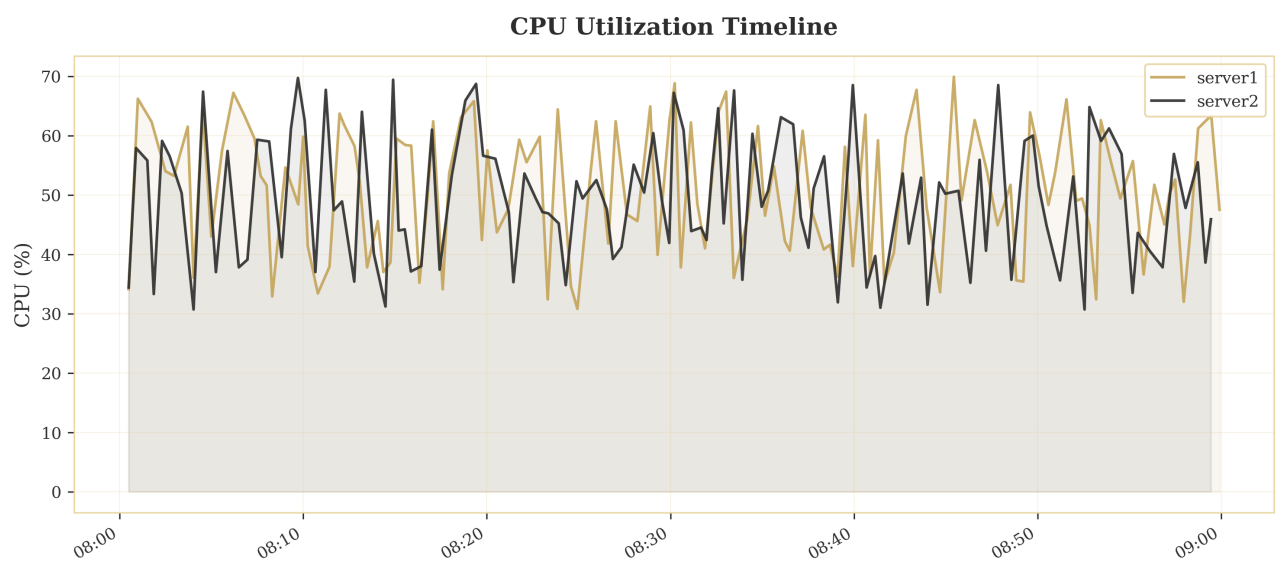


Figure 8: CPU Utilization Over Time

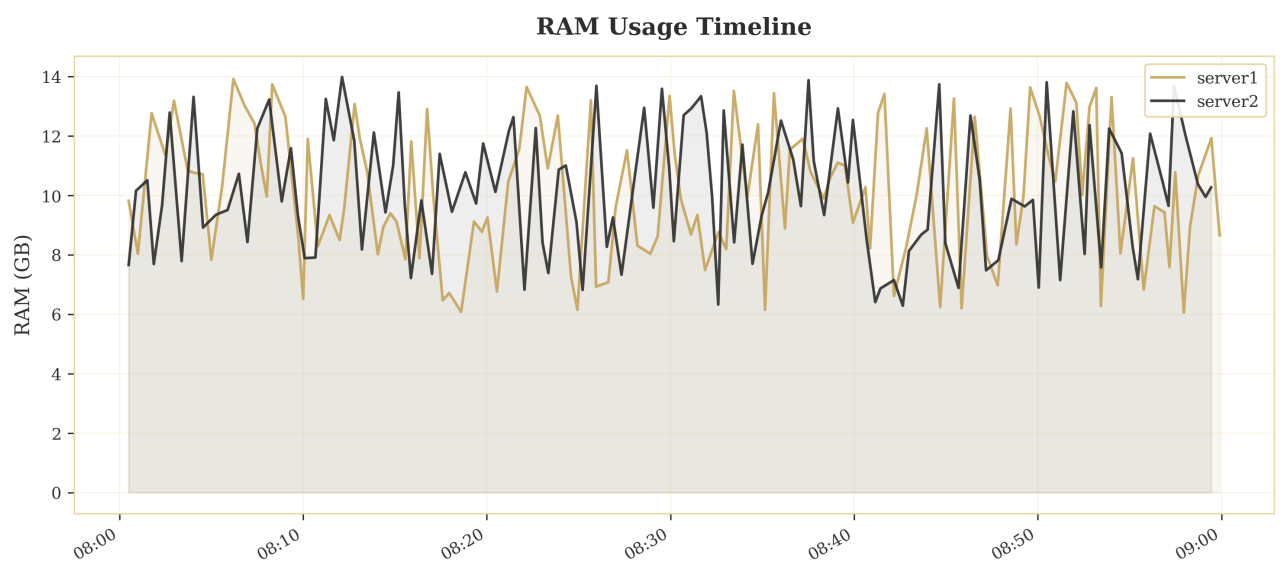


Figure 9: RAM Usage Over Time

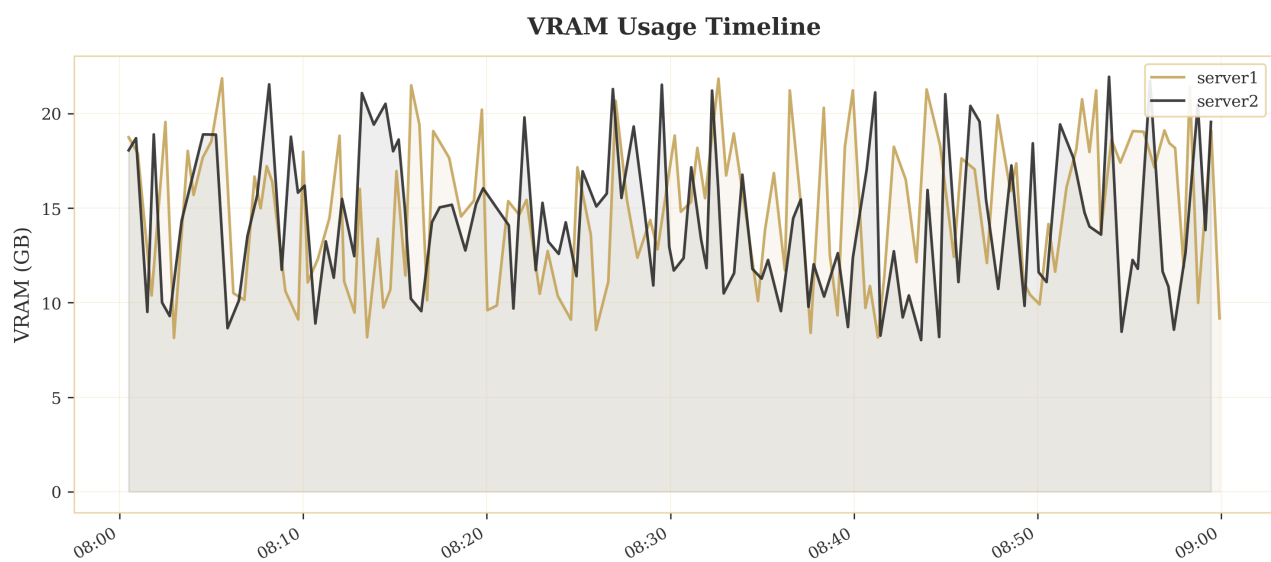


Figure 10: VRAM Usage Over Time

5 Detailed Breakdown

Performance metrics grouped by tool and test scenario.

Tool	Scenario	34.142.222.133				35.186.159.250			
		TPS	TTFT	P99	Err%	TPS	TTFT	P99	Err%
ollama_native	simple_chat	68.43	146.85	566.71	1%	57.62	359.58	436.66	0%
ollama_native	code_generation	63.15	265.15	623.98	0%	34.88	255.07	307.24	0%
ollama_native	long_context	58.69	73.13	320.16	1%	50.96	276.51	1123.40	1%
ollama_native	reasoning	55.67	120.45	503.40	5%	45.67	201.96	1197.12	2%
ollama_native	translation	48.02	152.14	1073.36	0%	31.95	348.12	289.15	1%
oha	simple_chat	70.40	182.58	335.79	0%	37.51	121.12	645.91	0%
oha	code_generation	25.06	285.89	396.31	2%	29.04	255.17	992.50	0%
oha	long_context	79.94	153.14	758.53	0%	37.11	206.86	734.37	1%
oha	reasoning	66.52	142.31	280.36	5%	58.38	300.99	223.43	0%
oha	translation	59.37	130.84	484.81	0%	19.28	112.98	725.56	0%
litellm	simple_chat	64.42	136.84	724.15	0%	55.50	161.69	725.65	0%
litellm	code_generation	54.25	108.02	965.88	0%	38.49	177.17	411.02	2%
litellm	long_context	31.83	121.59	637.02	1%	48.52	303.83	480.32	0%
litellm	reasoning	60.65	235.38	684.58	1%	48.19	132.08	465.94	0%
litellm	translation	65.54	174.03	1266.20	1%	43.64	330.00	545.16	1%

6 Environment & Integrity

6.1 Benchmark Environment

- **Platform:** aiDaptive Benchmark Suite v2.1
- **Execution Time:** 2026-05-06 08:00:00 – 2026-05-06 09:00:00
- **Prompt Set:** Default
- **Concurrency Levels:**

6.2 Integrity Verification

- **Raw Tool Output:** Captured & verified for each test case
- **Per-Prompt Logging:** Enabled (TTFT, TPS, TPOT per request)
- **Audit Package:** Available as ZIP with SHA-256 signed MANIFEST
- **Cryptographic Hash:** SHA-256 validation enabled

Note: This report is auto-generated from raw benchmark tool outputs. To verify the cryptographic integrity, download the Evidence Package (ZIP) which includes a MANIFEST.json with per-file SHA-256 hashes.

— End of Report —